	A	B	C	D
(a)	3	2	4	1
(b)	2	3	4	1
(c)	3	2	1	4
(d)	2	3	1	4

Ans. (b) : Wadhwa Commission was appointed for killing of Graham Staines, Liberhan Commission for Demolition of Babri Masjid at Ayodhya, Sri Krishna Commission appointed for riots in Mumbai and Jain Commission for assassination of Rajiv Gandhi.

134. Assume that the moon takes exactly 30 days to complete the cycle and also assume that it rises in the East exactly at 6.48 p.m. on the first day. On the fourth day, at what time will it rise?

(a) 8:24 p.m. (b) 9.12 p.m.
(c) 10.00 p.m. (d) 11.48 p.m.

Ans. (b) : Total time taken by the moon to complete the cycle = 30 days.

$\therefore$ consecutive moon rise takes a time of $\frac{24}{30}$

$$= \frac{20\times 60}{39}\text{minutes} = 48\,\text{minutes}$$

Now first day moon rise at 6 : 48 pm

Then 2nd day = 6 : 48 + 0 : 48 = 7 : 36

3rd = 7 : 36 + 0 : 48 = 7 : 84 = 8 : 24

4th = 8 : 24 + 0 : 48 = 8 : 72

= 9 : 12 pm

135. A complete cycle of the traffic light takes 60 seconds. During each cycle the light is green for 25 seconds, yellow for 5 seconds, and red for 30 seconds. At a randomly chosen time, the probability that the light will not be green is:

(a) 1/3 (b) 1/4
(c) 5/12 (d) 7/12

Ans. (d) : Probability that the light is not green

$$= \frac{\text{Time for which light is not Green}}{\text{Time taken for the entire cycle}}$$

$$= \frac{5+30}{60} = \frac{35}{60} = \frac{7}{12}$$

136. A solid cube just gets completely immersed in water when a 0.2 kg mass is placed on it. If the mass is removed, the cube is 2 cm above the water level. What is the length of each side of the cube?

(a) 12 cm (b) 10 cm
(c) 8 cm (d) 6 cm

Ans. (b) : Let the side of cube is x then the volume of the 2 cm part of the cube = $2 \times x \times x = 2x^2$

Mass displaced by the portion = 0.2 kg = 200 gm.

Now mass = volume × density $\Rightarrow 200 = 2x^2 \times 1 \Rightarrow x^2 = 100$ or $x = 10$ cm.

137. Who among the following won the men's single title at the World Badminton Championship in the year 2001?

(a) Gopichand (b) Hendrawan
(c) Ji Xin Peng (d) Peter Gade

Ans. (b) : Hendrawan of Indonesia won the men's singles title at the world Badminton Championship in the year 2001, defeating Peter Gade of Sweden in the final.

138. A trader fixed the price of an article in such a way that by giving a rebate of 10% on the price fixed the made a profit of 15%. If the cost of the article is Rs. 72, the price fixed on it is:

(a) Rs. 82.80 (b) Rs. 90.00
(c) Rs. 92.00 (d) Rs. 97.80

Ans. (c) : Let the fixed price of the article be x

then S.P. = 90% of $x = \frac{9x}{10}$

Also the S.P. is 115% of 72 $= \frac{115\times 72}{100}$

$$\Rightarrow \frac{9x}{10} = \frac{115\times 72}{100}$$

$$\Rightarrow x = \frac{115\times 72\times 10}{9\times 100} = 92$$

139. Consider the volumes of the following:

1. A parallelepiped of length 5 cm, breadth 3 cm and height 4 cm.
2. A cube of each side 4 cm.
3. A cylinder of radius 3 cm and length 3 cm.
4. A sphere of radius 3 cm.

The volumes of these in the decreasing order is

(a) 1, 3, 2, 4 (b) 4, 2, 3, 1
(c) 1, 2, 3, 4 (d) 4, 3, 2, 1

Ans. (d) : Volume of parallelepiped = $l \times b \times h$

$= 5 \times 3 \times 4 = 60\ \text{cm}^3$

Volume of cube = 4 × 4 × 4 = 64 cm3

Volume of cylinder = $\pi r^2 h = \pi \times 32 \times 3 = 84.7$ cm3

Volume of sphere $= \frac{4}{3}\pi r^3 = \frac{4}{3}\times\frac{22}{7}\times 3^3 = 113.04\,\text{cm}^2$

$\therefore$ required order is 4, 3, 2, 1.

140. A train of length 150 metres, moving at a speed of 90 km/hr can cross a 200 metre bridge in

(a) 8 seconds (b) 14 seconds
(c) 6 seconds (d) 15 seconds

Ans. (b) : Total distance to be crossed by the train

= 150 + 200 = 350 m

Speed = 90 km/hr $= \frac{90\times 1000}{3600}$ m/s = 25 m/s

Time taken $= \frac{350}{25} = 14$sec.

141. Match List I with List II and select the correct answer using the codes given below the lists:

	List-I (Book)		List-II (Author)
A.	Autobiography of unknown Indian	1.	Mulk Raj Anand
B.	India : A wounded civilization	2.	Nirad C. Chaudhari
C.	Confession of a lover	3.	R.K. Narayan
D.	The English Teacher	4.	V.S. Naipaul

	A	B	C	D
(a)	1	3	2	4
(b)	2	4	1	3
(c)	1	4	2	3
(d)	2	3	1	4

Ans. (b) :

	List-I (Book)		List-II (Author)
A.	Autobiography of unknown Indian	1.	Mulk Raj Anand
B.	India : A wounded civilization	2.	Nirad C. Chaudhari
C.	Confession of a lover	3.	R.K. Narayan
D.	The English Teacher	4.	V.S. Naipaul

142. Two ladies simultaneously leave cities A and B connected by a straight road and travel towards each other. The first lady travels 2 km/hr faster than the second lady and reaches B one hour before the second lady reaches A. The two cities A and Bare 24 km apart. How many kilometers does each lady travel in one hour?

(a) 5 km, 3 km (b) 7 km, 5 km
(c) 8 km, 6 km (d) 16 km, 14 km

Ans. (c) : Let the speed of 2nd lady be x km/hr

Then speed of 1st lady = $(x + 2)$ km/hr

$$\therefore \frac{24}{x} - \frac{24}{(x+2)} = 1$$

$$\Rightarrow x(x+2) = 48$$

$$\Rightarrow x^2 + 2x - 48 = 0$$

$$\Rightarrow x^2 + 8x - 6x - 48 = 0$$

$$x(x+8) - 6(x+8) = 0$$

$$(x+8)(x-6) = 0$$

Neglecting the negative value of x

$x = 6$

Hence speed of 1st lady 6 + 2 = 8 km/hr

and speed of 2nd lady = 6 km/hr

143. Amit started a business by investing Rs. 30,000. Rahul joined the business after some time and invested Rs. 20,000. At the end of the year, profit was divided in the ratio of 2 : 1. After how many months did Rahul join the Business?

(a) 2 (b) 3
(c) 4 (d) 5

Ans. (b) : Let after t months Rahul joined the business

$\because$ Amit does the business for 1 year and Rahul for (12 – t) months

$\therefore$ They will share the profit in ratio

$30000 \times 12 : 20000 \times (12 - t) = 2 : 1$

$$\frac{360000}{24000 - 20000t} = \frac{2}{1}$$

$40000\,t = 48000 - 36000$

$= 120000$

$$t = \frac{12000}{40000} = 3\text{months}$$

144. When the time in the wall-clock is 3.25 p.m. the acute angle between the hours-hand and the minutes-hand is

(a) 60º (b) $52\frac{1}{2}°$
(c) $47\frac{1}{2}°$ (d) 42º

Ans. (c) : In a clock, the angle between two successive number is $\frac{360}{12} = 30°$

When the time is 3.25 pm, the minute hand will be on 5 and will have moved 60° from 3 and hour hand would be between 3 and 4 and as it moves 30° in 60 minutes, so in 25 minutes it would move

$$\frac{30°\times 25}{60} = 12.5$$

So the angle between the two hands

$$60° - 12.5° = 47.5 = 47\frac{1}{2}^\circ$$